

COVID-19 Research Paper Analyzer

Automated Knowledge Extraction and Classification using NLP

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Abstract

The exponential growth of biomedical literature during the COVID-19 pandemic created an "infodemic," making it difficult for medical professionals to extract actionable insights. This project develops an end-to-end Natural Language Processing (NLP) pipeline to analyze 10,000 medical abstracts from the CORD-19 dataset. By integrating NLTK, spaCy, and Gensim, we performed linguistic preprocessing, topic modeling, semantic mapping, and classification. We implemented both classic Machine Learning (Random Forest) and Deep Learning (Bi-LSTM) to categorize papers into treatment-focused versus public-health-focused research. Ultimately, the Random Forest model outperformed the Bi-LSTM due to the frequency-based nature of the target variables. This report details the methodology, results, and a critical comparison of the NLP tools utilized.

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1 Introduction

1.1 Problem Statement from a Healthcare Perspective

During global health crises, the rapid publication of scientific literature is both an asset and a liability. Throughout the SARS-CoV-2 pandemic, thousands of papers were published weekly. For clinicians and researchers, manually reviewing abstracts to identify critical clinical trials, treatment efficacies, or epidemiological data is an insurmountable task. There is a critical need for automated systems capable of filtering, categorizing, and summarizing massive biomedical datasets.

1.2 Why Automated Knowledge Extraction Matters

Automated knowledge extraction acts as a triage system for scientific literature. By leveraging Natural Language Processing (NLP) and Machine Learning (ML), we can instantly extract Named Entities (e.g., diseases, chemical compounds, geographic locations), discover hidden thematic clusters, and map semantic relationships between novel medical terms. This accelerates evidence-based medicine and policy-making.

1.3 Research Questions Addressed

This study addresses the following questions:

- How can unsupervised topic modeling reveal the underlying research themes in the CORD-19 dataset?
- How do standard ML models (Random Forest) compare against advanced sequential DL models (Bi-LSTM) when classifying synthetically labeled medical abstracts?
- What are the comparative strengths and trade-offs of NLTK, spaCy, and Gensim in a biomedical NLP pipeline?

2 Dataset & Exploratory Data Analysis (EDA)

2.1 CORD-19 Dataset Description

The dataset utilized is a subset of the COVID-19 Open Research Dataset Challenge (CORD-19), consisting of 10,000 raw medical abstracts. The data includes heterogeneous text ranging from highly technical molecular biology papers to broad epidemiological studies.

2.2 Key EDA Findings

Initial Exploratory Data Analysis revealed a heavy right-skew in abstract lengths, with most abstracts containing between 150 and 300 words. Frequency distributions highlighted that standard English stopwords masked the clinical vocabulary. After initial tokenization, the most frequent domain-specific words included *patients*, *virus*, *disease*, *severe*, and *treatment*. A WordCloud visualization confirmed that the dataset was heavily skewed toward symptomatic outcomes and viral transmission.

2.3 Data Quality Analysis

The raw text contained significant noise, including HTML tags, digital object identifiers (DOIs), unstandardized mathematical symbols, and redundant academic phrasing (e.g., "In this study," "We conclude that"). This necessitated a rigorous, multi-staged preprocessing pipeline.

3 Methodology

3.1 Phase 1: NLTK Preprocessing Decisions

The Natural Language Toolkit (NLTK) was employed for foundational text sanitization. The text was lowercased, and regular expressions (Regex) were used to strip punctuation, numerical artifacts, and special characters. We expanded the default NLTK English stopword list to include custom academic stopwords (e.g., *study*, *paper*, *result*, *analysis*) to prevent these high-frequency, low-information words from dominating the predictive models.

3.2 Phase 2: spaCy NLP Choices

For advanced linguistic parsing, we utilized **spaCy**. Unlike NLTK's string-based approach, spaCy creates object-oriented documents. We performed:

- **Lemmatization:** Reducing words to their base morphological form (e.g., *patients* to *patient*).
- **Part-of-Speech (POS) Tagging:** Isolating Nouns and Adjectives to capture the descriptive essence of the abstracts.
- **Named Entity Recognition (NER):** Highlighting specific entities such as *ORG* (organizations like WHO), *PRODUCT* (drugs), and *GPE* (geopolitical entities).

3.3 Phase 3: Gensim LDA + Word2Vec Design

Gensim was utilized for unsupervised learning. To discover latent research themes, we trained a Latent Dirichlet Allocation (LDA) model. Concurrently, a Word2Vec model

was trained using the Skip-Gram architecture to map words into continuous vector space, allowing the system to understand that terms like *Remdesivir* and *antiviral* appear in similar semantic contexts.

4 Results: Topic Modeling

4.1 Discovered Topics

The LDA model successfully partitioned the corpus into distinct research themes. The optimal number of topics was determined to be 7, achieving a strong coherence score. Key topics included:

1. **Clinical Trials:** *vaccine, dose, efficacy, placebo, trial*
2. **Public Health/Epidemiology:** *lockdown, transmission, mask, social, distancing*
3. **Molecular Biology:** *protein, rna, genome, spike, binding*

4.2 Word2Vec Semantic Findings

The Word2Vec embeddings successfully captured biomedical semantics. For instance, querying the most similar words to *vaccine* yielded *mRNA, immunization, dose, and booster* with similarity scores exceeding 0.75. This proves the model learned the contextual meaning of medical terms purely from their co-occurrence.

5 Results: Classification

To evaluate predictive capabilities, a synthetic rule-based target was generated: abstracts containing treatment-related keywords were labeled ‘1’ (Treatment & Clinical Trials), and others ‘0’ (Other).

5.1 ML vs DL Model Comparison

We compared a classic Random Forest Classifier (using TF-IDF vectorization) against a Bidirectional Long Short-Term Memory network (Bi-LSTM, using sequential tokenization and embeddings).

Model	Accuracy	F1-Score
Random Forest (TF-IDF)	0.945	0.921
Bi-LSTM (Sequential)	0.882	0.865

Table 1: Classification Metrics on Test Set

5.2 Error Analysis Findings

Surprisingly, the Random Forest model outperformed the Bi-LSTM. Error analysis revealed this is highly logical: because the ground-truth labels were generated using rule-based keyword matching, the TF-IDF vectorizer—which explicitly maps word frequencies—provided the most direct mathematical signal. The Bi-LSTM, designed to capture complex grammatical sequencing and context, overfit the data and attempted to find complex patterns where only simple frequency patterns existed.

6 Critical Tool Comparison

6.1 NLTK vs. spaCy vs. Gensim

- **NLTK:** Highly effective for basic, low-level string manipulation (tokenization, stop-words). It is computationally lightweight but lacks deep linguistic understanding. It treats text as raw data.
- **spaCy:** The industry standard for production-level NLP. It excels at complex, context-aware tasks like Dependency Parsing and NER. However, it is computationally heavy and slower than NLTK for simple tasks.
- **Gensim:** Operates on an entirely different paradigm. While NLTK and spaCy focus on grammar and syntax, Gensim focuses on *semantics and mathematics* (Topic Modeling, Vector Embeddings). It is the premier tool for discovering unsupervised meaning in large corpora.

6.2 Recommendation for Biomedical NLP

For biomedical text pipelines, a hybrid approach is mandatory. NLTK should be used for initial rapid cleaning; spaCy (or specifically **scispaCy**) should be utilized for extracting clinical entities; and Gensim must be applied to map the semantic relationships between novel viruses and experimental drugs.

7 Conclusion

7.1 Key Findings Summary

This project successfully developed an automated NLP pipeline capable of analyzing 10,000 COVID-19 research papers. We demonstrated that integrating NLTK, spaCy, and Gensim extracts deep semantic value from unstructured text. Furthermore, our classification results proved a fundamental Data Science axiom: advanced Deep Learning (Bi-LSTM) is not universally superior to classic Machine Learning (Random Forest) if the underlying data patterns are frequency-based rather than sequence-based.

7.2 Limitations

The primary limitation of this study is the reliance on a synthetically generated, rule-based target variable for classification. While useful for pipeline validation, it creates an artificial ceiling on model complexity. Additionally, analyzing abstracts rather than full-text articles omits granular methodology and specific statistical results.

7.3 Future Work

Future iterations should utilize manually annotated datasets to train the classification models, allowing the Bi-LSTM to reach its full potential in understanding clinical context. Finally, deploying this backend architecture via a modern React/FastAPI web interface will transition this research from a Jupyter Notebook pipeline into a production-ready medical software tool.